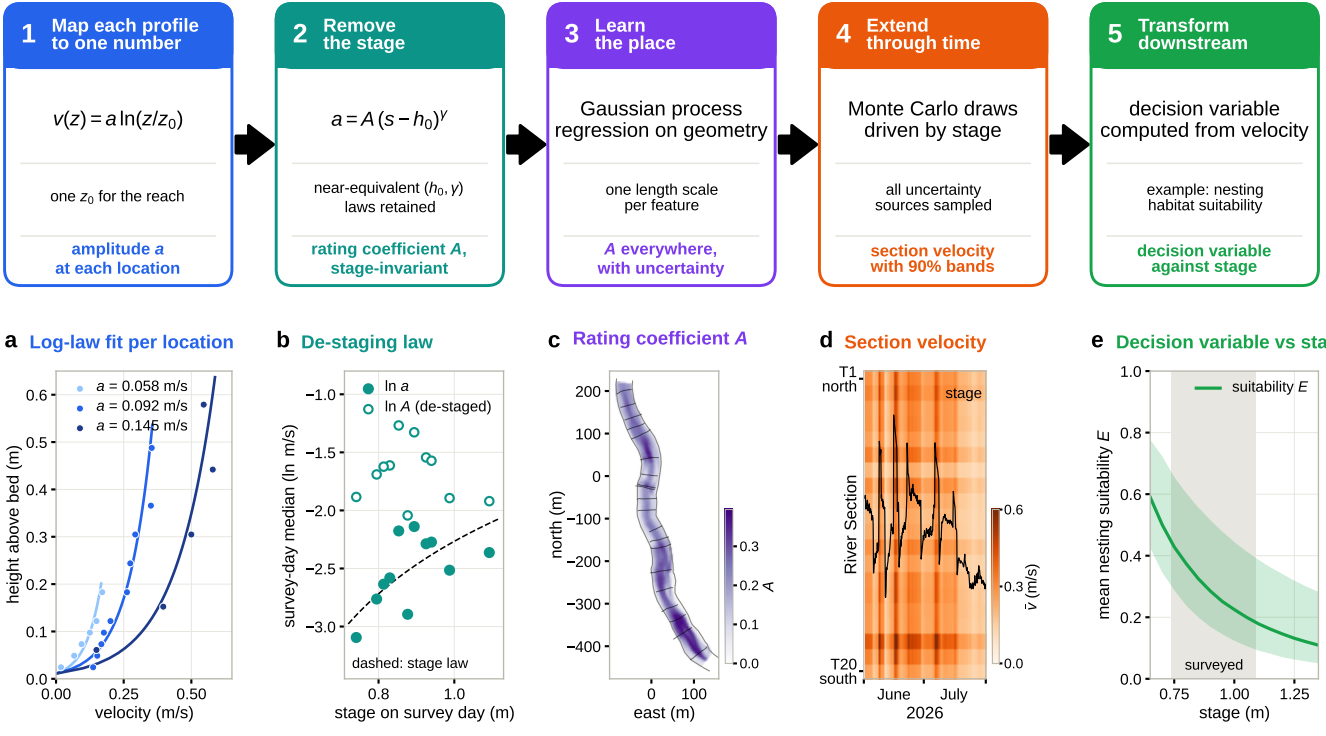

1 Map each profile to one number
$v(z) = a \ln(z/z_0)$
one z_0 for the reach
amplitude a at each location
2 Remove the stage
$a = A\,(s - h_0)^{\gamma}$
near-equivalent (h_0, γ) laws retained
rating coefficient A, stage-invariant
3 Learn the place
Gaussian process regression on geometry
one length scale per feature
A everywhere, with uncertainty
4 Extend through time
Monte Carlo draws driven by stage
all uncertainty sources sampled
section velocity with 90% bands
5 Transform downstream
decision variable computed from velocity
example: nesting habitat suitability
decision variable against stage
a Log-law fit per location
a = 0.058 m/s
a = 0.092 m/s
a = 0.145 m/s
height above bed (m)
velocity (m/s)
b De-staging law
ln a
ln A (de-staged)
dashed: stage law
survey-day median (ln m/s)
stage on survey day (m)
c Rating coefficient A
north (m)
east (m)
A
d Section velocity
T1 north
T20 south
stage
River Section
June
July
2026
$\bar{v}$ (m/s)
e Decision variable vs stage
suitability E
surveyed
mean nesting suitability E
stage (m)